SAMPLE

Famous Astronomers: Hipparchus

The Father of Trigonometry

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Introduction

Hipparchus of Nicaea, who lived from roughly 190 to 120 before the common era, is the astronomer the ancients themselves ranked first, and the verdict has survived twenty-two centuries of revisers. He worked mostly from the island of Rhodes, under the same stars that the sailors of the Aegean had steered by for a thousand years, and out of that ordinary Mediterranean sky he built the first mathematical astronomy: models that predicted the Sun and the Moon, a catalog that fixed the stars, tables that turned angles into numbers, and a discovery, the slow wobble of the heavens now called precession, that no one before him had even imagined looking for.

The man is almost entirely lost, and this is the paradox his biography must live with. One thin book of his survives, a commentary on a poem about the constellations, a work he wrote young and apparently dismissed himself; everything else, the great treatises on the Sun and the Moon, the star catalog, the chord tables, the attack on Eratosthenes' geography, exists only as fragments and secondhand reports, quoted by Ptolemy four centuries later, paraphrased by Strabo, praised by Pliny, and reconstructed by modern scholars the way paleontologists rebuild a skeleton from scattered bones. What survives of Hipparchus is mostly what other people said he did, and what he did, on that testimony, was everything.

He is called the founder of trigonometry, the discoverer of precession, the inventor of the star catalog and the brightness scale, the measurer of the Moon's distance, the improver of eclipse prediction, and the first astronomer whose work we can follow as a chain of observation, mathematics, and honest self-correction. Ptolemy, who inherited the structure of the Almagest from him, called him a lover of truth in the arithmetic of the heavens, and the modern verdict of the historians is the ancient one restated: between the stargazers of the Babylonian temples and the mathematicians of the Renaissance, no single mind did more to make astronomy a science.

The chapters ahead tell him in order: the meager life and the rich aftermath, the method that joined Babylonian records to Greek geometry, the solar and lunar theories, the instruments and the catalog, the discovery of precession, the lost works and the scholarly war over what Ptolemy preserved, the modern reconstructions and the unexpected resurrections, from a bronze mechanism in a shipwreck to a satellite named in his honor. A biography that is mostly a detective story, the reader is warned at the start, and the detective work, in this case, is the man's own legacy: he wrote everything to be checked, and the checking continues.

The name's own spelling wobbles in the sources, Hipparchus and Hipparchos both in the modern literature, and the wobble is a fair emblem of the evidence's texture: even the spelling rests on inference, the Greek and the Latinized forms trading places across the manuscripts and the centuries. The reader will find this volume using the Latin form, as the series does, but should know the original lies one letter-shape away, that his own language's alphabet wrote him with a character this book's pages keep in its notes rather than its lines, and that the man's name has been pronounced differently in every century that honored it.

A word on the volume's evidence discipline, stated at the outset so that later chapters need not pause for it. Everything here rests on three tiers: the surviving text, the ancient quotation, and the modern recomputation; where the tiers agree, the account states a fact, where they diverge, the account states the divergence, and where the gap is silence, the account says unknown. The reader who checks the scholarly editions behind any paragraph will find more argument than a popular volume can carry, and the biography's honesty is the same habit the subject's was: the errors and the uncertainties are reported, not smoothed, because a smoothed Hipparchus would be a false one, and he spent his life against false numbers.

Chapter 1: The Discovery of Hipparchus

The discovery of Hipparchus, in the biographical sense, begins with how little there is to discover. He was born, the ancient evidence agrees, in Nicaea in Bithynia, the region of what is now northern Turkey, and the city was proud enough of him to strike coins in his honor centuries later, showing his face beside a globe; he worked, the datings of his observations indicate, mostly at Rhodes; he is recorded in at least one place as present in Alexandria; and he died, by the reckoning of his birth and last observations, around 120 before the common era. That sentence, minus the coins, is nearly the whole documentary life, and the biographer's first lesson is that antiquity's greatest observational astronomer left the world no letters, no anecdotes worth trusting, and no contemporary portrait.

The dates hold on the astronomy itself, and the chapter explains how. Hipparchus's observations, as reported by Ptolemy and other sources, run in a narrow band: the earliest datable records cluster in the 160s and 140s before the common era, the latest in 127 and 126, and his own references to earlier observers, to Timocharis three

centuries before him, to the Babylonian archives, set the boundaries of his data. A man whose working life is visible as a set of dated observations, rather than as a set of dated events, is a special kind of historical figure, and the biographer's honest conclusion is that we know Hipparchus's career the way we know a star's motion: not from watching him, but from plotting the fixed points he left.

The survival question shapes everything else, and it deserves the chapter's centerpiece paragraph. Antiquity's greatest astronomer is preserved in one book, the Commentary on the Phaenomena, his critique of the sky descriptions of Eudoxus and Aratus, a work of his earlier years that survives because it was readable, teachable, and never stopped being copied; the great works are gone, and the loss is the civilization's, not his. What he wrote on the Sun and the Moon, his star catalog and the maps that carried it, his three books against Eratosthenes' geography, his chord tables in twelve books, all perished with the libraries and the slow decay of the technical manuscript tradition, and the modern reader knows Hipparchus the way his own posterity did, through the summary Ptolemy made and the fragments the quoting authors preserved.

The discovery chapter closes with the man's own temperament, which the sources do preserve, sideways. He was, by every testimony, a debunker: the commentary that survives is a list of other people's errors, patiently corrected; the geography treatise was an attack on the most famous geographer of the age; the eclipse work was an audit of Babylonian predictions improved; and even his own results are reported with their error bars attached, with candid notes on what he could not resolve. The biographical verdict, on evidence this thin, is the only safe one: Hipparchus was the first astronomer whose personality survives as a method, skeptical, quantifying, and constitutionally unable to let a wrong number stand, and the twenty centuries of checking that followed have kept finding him right.

The social world of the life deserves its paragraph, thin as its record is. He worked in the Hellenistic world's long second-century before the common era, the era between the great kingdoms' wars, when Rhodes was the merchant republic of the eastern Mediterranean, the friend of Rome and the port of the grain ships, wealthy, maritime, and obsessed with accurate time; the fitting context is that the man who fixed the year's length and the stars' places worked from an island whose living depended on navigation, on calendars, and on the exact knowledge of winds and seasons, and the seafaring city's practical demands were, plausibly, the young astronomer's first clients, whatever his own motives.

The Alexandria presence is the one biographical detail with a source, and the chapter should state it carefully: a papyrus fragment, and the later doxography, associate Hipparchus with the Egyptian capital's scientific circles, and his solar observations are reported from latitudes compatible with both Alexandria and Rhodes, so the reasonable picture is a working career with stations in both cities, using the library and the instrument-makers of the one from the observing platforms of the other; the detail matters because it shows the program's scale, a science networked from the start, and it prepares the geography chapter's point that the Mediterranean itself was one of his instruments.

The chronology of the career deserves a closing paragraph in the discovery chapter, because the dated observations sketch a working life. The earliest datable activity falls in the middle decades of the second century before the common era, in the 160s and 150s by the count, when he was comparing the received records and beginning his own solstice work; the great decade runs from roughly 147 to 127, the years of the lunar program, the star catalog's making, and the precession comparisons; and the last securely dated observations, from 127 and 126, are the ones Ptolemy quotes as the standard. The sketch shows a career of roughly four working decades, the middle ones the productive core, and the biographer's note is the shape's modernity: a research program with phases, a literature with a peak, and a retirement into sources the record simply does not preserve.

The discovery chapter can end with the afterlife of his fame, because the ancients' verdict has a documentary trail. Ptolemy, four centuries on, cites him more than any other source and defers to his observations as the standard; Pliny, writing the Roman world's encyclopedia of nature, devotes to him the famous passage on the new star and the catalog, calling the work a bequest to mankind; Strabo, the geographer, carries his fragments as the sharp edge of the science; and the later compilers, from Theon to the Byzantines, pass his name down the centuries as the founder's. The trail's meaning for the chapter is that the thin biography sits on a heavy reputation, and the reputation, unlike the works, was never lost, which is the rarest bequest of all, a name that survived on merit rather than on anecdote.